

Achieved ceiling on the standard axes

MAL-BIND-K-CT $\wedge$
MAL-BIND-K-PK (+ K-CTX)

ContextBound (this work)

X-Wing (seed-dk)

*both reach it — we are **not** “stronger” here*

$\supseteq$ LEAK-BIND $\supseteq$ HON-BIND (CDM monotonicity)

X-BIND-CT-*

(ciphertext binds K/pk)

× structurally **unachievable** for implicitly-rejecting
KEMs — *not claimed*

ContextBound: $\leq$ CR(SHA3) *alone* — zero KEM assumption, machine-checked (EasyCrypt, CDM Fig. 6).

X-Wing: relies on ML-KEM FO self-binding + the seed-dk format (CCR proven; MAL by draft).

Honest position: same ceiling as X-Wing on the standard $\{\text{CT}, \text{PK}\}$ axes; our edge is *assumption-minimality + mechanization + dk-format robustness*, **not** a stronger notion.